

FAIRFIELD, IA, USA

WMO: 726498

Lat: 41.053N

Lon: 91.979W

Elev: 244

StdP: 98.43

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.0	(c) -17.2	(d) -24.2	(e) 0.4	(f) -19.2	(g) -22.0	(h) 0.5	(i) -16.4	(j) 13.1	(k) -8.1	(l) 12.0	(m) -5.9	(n) 3.9	(o) 300	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.0	25.3	32.6	24.8	31.3	24.0	27.0	32.4	25.9	30.9	25.0	29.9	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.7	31.0	24.2	19.7	29.3	23.0	18.3	28.3	86.9	32.5	81.9	30.7	77.1	29.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.9	8.7	-23.7	36.5	3.6	2.4	-26.3	38.2	-28.4	39.6	-30.4	40.9	-33.1	42.7
			-24.4	28.5	3.8	1.0	-27.2	29.2	-29.4	29.8	-31.5	30.4	-34.3	31.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-4.2	-2.7	4.4	11.2	17.3	22.5	24.4	23.0	19.1	12.0	5.1	-1.7		
	DBStd	11.42	6.81	6.83	6.54	5.63	4.74	3.47	3.38	3.25	4.59	5.36	5.91	6.56		
	HDD10.0	1619	441	357	196	51	3	0	0	0	1	39	167	364		
	HDD18.3	3298	699	588	433	221	78	7	1	3	45	206	397	621		
	CDD10.0	1962	0	2	23	88	229	373	446	404	273	103	20	2		
	CDD18.3	601	0	0	2	8	46	130	189	149	67	11	1	0		
	CDH23.3	5863	0	0	11	106	428	1218	1973	1364	653	107	3	0		
	CDH26.7	2204	0	0	1	23	125	448	844	514	227	23	0	0		
(14) Wind	WSAvg	4.1	4.6	4.6	4.9	5.1	4.4	3.6	2.9	2.7	3.2	3.9	4.5	4.4		
(15) Precipitation	PrecAvg	951	37	31	63	92	119	103	123	105	101	78	57	42		
	PrecMax	1452	90	117	142	193	315	212	370	265	278	241	162	146		
	PrecMin	497	2	4	7	17	35	22	5	23	3	1	0	2		
	PrecStd	199	24	22	39	42	63	53	86	57	63	54	38	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	17.7	25.1	29.4	32.7	34.1	37.3	36.1	33.8	29.1	23.7	17.3		
		MCWB	8.6	11.3	16.9	19.5	22.9	25.3	25.7	23.6	24.1	21.1	15.9	14.0		
	2%	DB	10.2	12.8	21.1	26.3	30.0	32.6	34.9	33.0	32.0	26.2	20.0	12.7		
		MCWB	7.0	8.6	15.2	17.0	21.7	24.6	26.5	25.4	22.9	18.2	13.9	9.9		
	5%	DB	7.2	9.1	17.5	23.7	27.6	31.3	32.7	31.4	29.1	23.7	17.2	9.8		
		MCWB	4.5	6.2	12.9	15.8	19.9	23.7	25.4	24.8	22.1	17.1	12.5	7.2		
	10%	DB	4.0	6.3	14.1	21.2	25.9	29.2	31.2	29.7	27.3	21.1	14.1	7.1		
		MCWB	1.9	3.8	9.6	14.1	18.8	22.6	24.5	23.7	20.3	15.9	9.9	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.7	18.5	20.5	24.7	28.0	28.6	28.3	25.8	22.7	17.3	15.1		
		MCDB	11.9	17.0	23.7	26.8	30.3	32.6	34.5	33.1	31.2	28.3	22.4	17.2		
	2%	WB	6.8	9.3	15.7	18.6	22.8	25.8	27.2	26.8	24.3	20.3	15.0	10.9		
		MCDB	9.7	13.0	19.3	25.1	28.2	30.7	33.1	31.6	29.7	24.0	18.6	12.4		
	5%	WB	4.4	6.4	13.3	16.8	21.4	24.9	26.2	25.5	23.0	18.3	12.9	7.4		
		MCDB	7.2	9.2	16.9	22.2	26.5	29.8	31.5	30.2	28.0	22.1	16.7	9.1		
	10%	WB	2.3	3.9	10.1	15.1	19.9	23.6	25.2	24.2	21.8	16.3	10.3	4.6		
		MCDB	4.1	6.2	14.1	20.3	24.4	28.2	30.0	28.3	26.0	20.6	14.0	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.3	10.6	12.2	10.9	10.6	11.0	11.1	12.5	11.6	10.3	8.5		
		MCDBR	11.9	13.3	14.5	15.6	13.0	12.1	12.1	12.7	13.4	14.3	13.7	11.3		
	5% WB	MCWBR	9.1	9.6	9.3	8.9	7.0	6.3	6.1	6.9	6.7	7.8	9.0	8.9		
		MCWBR	11.9	12.5	13.1	13.8	11.4	10.8	10.9	11.1	11.4	11.7	12.1	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.307	0.337	0.385	0.418	0.420	0.440	0.423	0.389	0.343	0.300	0.290		
	taud		2.421	2.384	2.392	2.266	2.244	2.284	2.215	2.279	2.324	2.456	2.508	2.451		
	Ebn at Noon		864	905	914	885	861	857	835	839	845	853	855	840		
	Edh at Noon		79	96	107	130	137	132	140	127	114	89	73	71		
(44) All-Sky Solar Radiation	RadAvg		1.87	2.84	3.75	4.75	5.57	6.19	6.20	5.44	4.66	3.10	2.09	1.57		
	RadStd		0.20	0.23	0.30	0.36	0.40	0.61	0.43	0.31	0.34	0.35	0.17	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page